Appendix: AI Usage Disclosure

A.1 Purpose of AI Assistance

Artificial Intelligence (AI) tools were used in this project to support development, documentation, and refinement of the system. The use of AI was limited to assistance in implementation efficiency and clarity of presentation, while all core design, modeling decisions, and system architecture were independently developed.

A.2 Scope of AI Usage

AI assistance was used in the following areas:

1. Code Assistance

- Generating boilerplate code for:
  - API structure (e.g., FastAPI endpoints)
  - Data preprocessing templates
  - Utility functions
- Debugging support for:
  - environment setup issues
  - dependency conflicts
  - runtime errors

All generated code was reviewed, modified, and integrated manually.

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2. Documentation and Writing

- Refinement of report sections for clarity and structure
- Formatting of technical explanations
- Assistance in drafting user guides and appendices

All content was validated and adapted to reflect the actual implementation.

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3. Conceptual Clarification

- Understanding MLOps tools and workflows (pipeline design, monitoring, orchestration)
- Clarification of theoretical concepts such as:
    - data drift detection
    - experiment tracking
    - containerization strategies

Final design decisions were made independently based on project requirements.

A.3 Areas Developed Independently

The following components were fully designed and implemented without direct AI generation:

- Problem formulation and system design
- Data preprocessing and feature engineering pipeline
- Feature selection methodology and leakage prevention
- Model training, evaluation, and comparison
- Integration of DVC, MLflow, and Airflow
- Monitoring system design (Prometheus + Grafana)
- UI workflow design and operations console logic
- Drift detection strategy and implementation
- Overall architecture and deployment pipeline

A.4 Compliance with Academic Guidelines

AI tools were used in accordance with the project guidelines:

- AI was not used to generate core logic without understanding
- All generated content was reviewed and modified
- No “zero-effort” or blind code usage was performed
- The student retains full ownership of system design and implementation

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A.5 Summary

AI tools were used as a **supporting aid** to improve productivity and clarity, but the **core intellectual contribution, system design, and implementation decisions remain the work of the student**.
